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Single-Channel EEG Brain–Computer Interface

From ECG Validation to P300 Cognitive Detection

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Abstract

A $100 single-channel bio-amplifier — the Backyard Brains SpikerBox — is paired with a custom Python pipeline to test how far low-cost neural sensing can go. Bypassing the vendor's display-only software, a direct binary serial decoder unlocks the raw signal and drives eight experimental paradigms: real-time ECG diagnostics (heart rate, variability, QTc); detection of the P300 "recognition" brain wave via an oddball task; a P300-based Concealed Information Test that works like a brain-based polygraph; demographically balanced facial-recognition lineups; a deliberate SSVEP test that maps the limits of a single frontal electrode; eye-blink and jaw-clench control of a computer game from electrical artifacts; and a $190 wearable neuro-prosthetic design for stroke rehabilitation. Across eleven experiments, seven succeeded, one was partial, and two informative failures each produced a reusable design rule. A 6.2 μV P300 was isolated from one electrode, and all 25 engineering requirements were verified. The central finding: high-level neural interfacing is constrained by algorithmic ingenuity, not hardware cost.

$100
sensor cost vs. $5,000+ clinical systems

6.2 μV
P300 recognition wave from ONE electrode

11
experiments — 7 passed, 2 instructive fails

25 / 25
engineering requirements verified

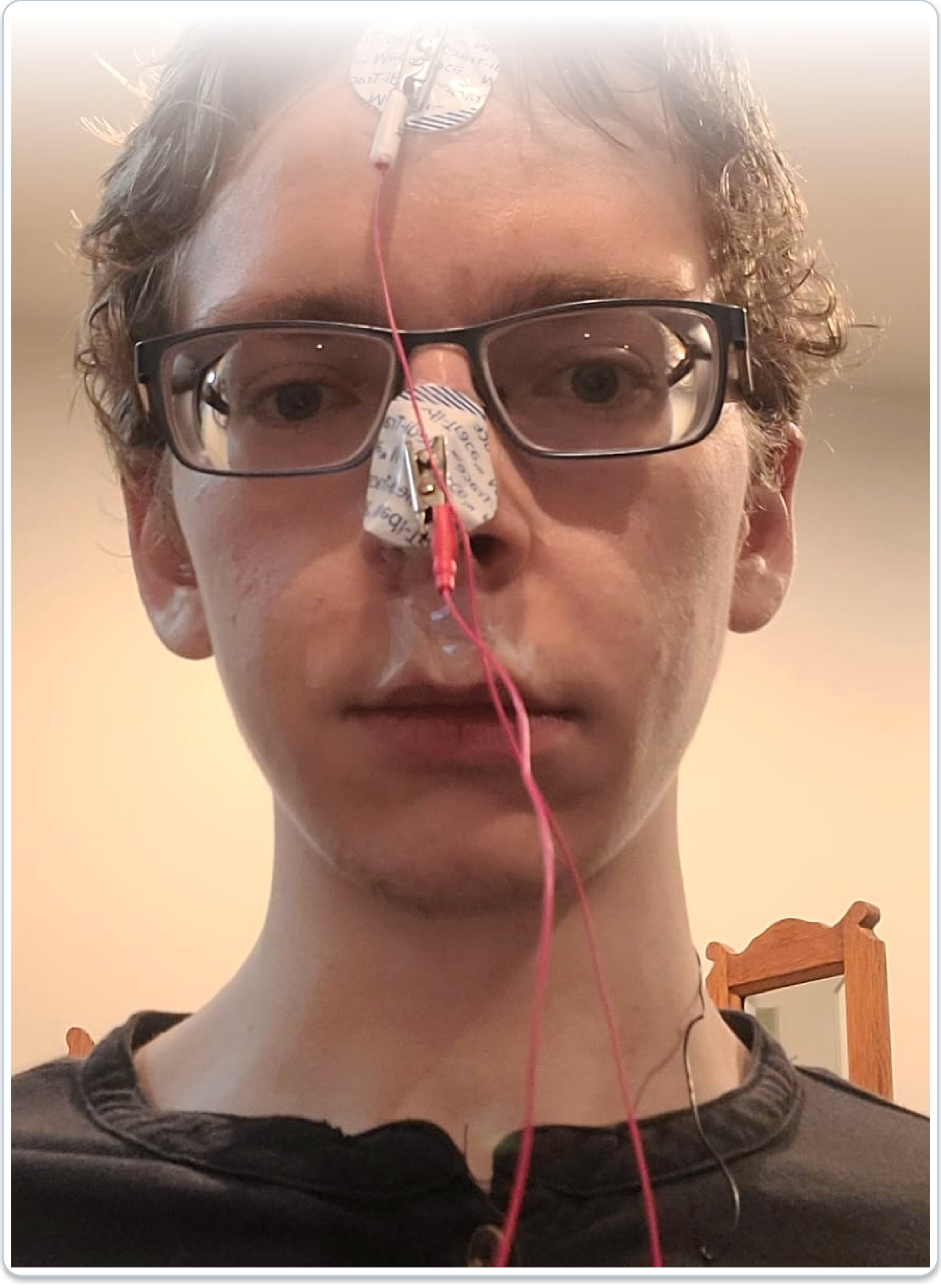
Motivation

Commercial brain–computer interfaces and clinical EEG systems cost thousands of dollars, placing neural sensing out of reach for students, makers, and small clinics. This project asks a blunt question: how far can a single $100 electrode go with the right software?

Objectives

- Unlock raw data from a $100 bio-amplifier with a custom serial decoder
- Detect and validate the P300 cognitive brain wave in real time
- Apply P300 to a forensic concealed-information test and face lineup
- Turn eye-blink / jaw-clench artifacts into hands-free device control
- Specify a $190 wearable neuro-prosthetic for stroke rehabilitation
- Verify all 25 engineering requirements

Experimental Setup



Single frontal (Fpz) electrode with a mastoid reference. Software event-markers — not push-buttons — avoid muscle-artifact contamination of the μV-scale signal.

Equipment & Software

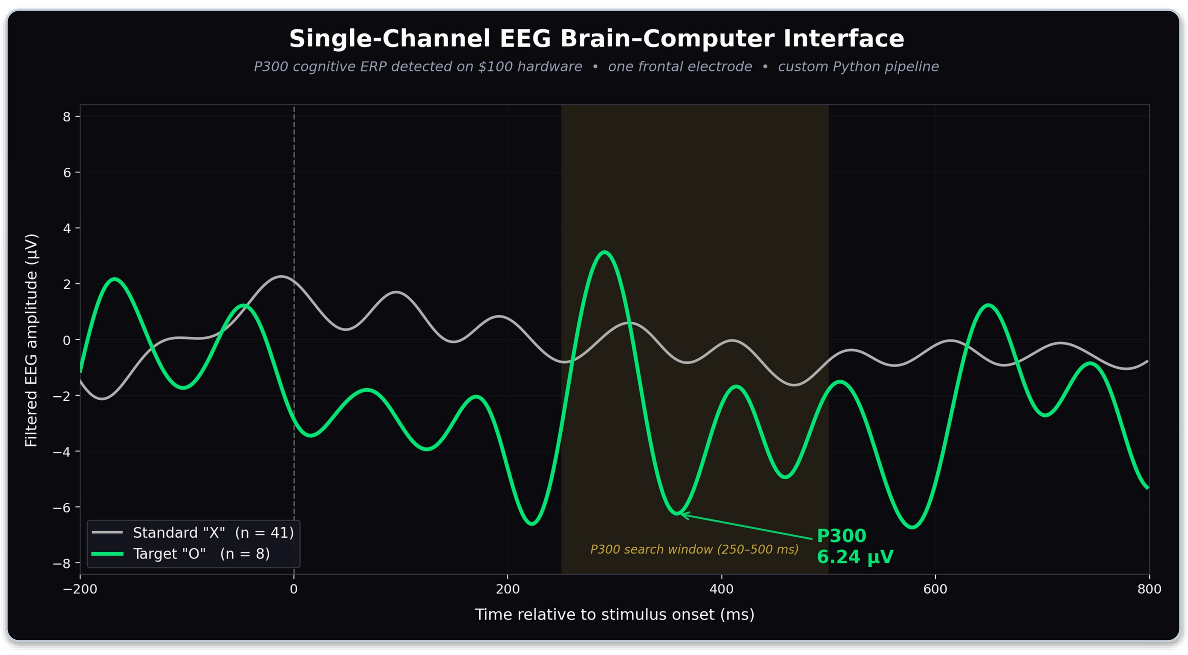
- Backyard Brains SpikerBox Heart & Brain analog front end (~$100)
- Medi-Trace Ag/AgCl pre-gelled electrodes (ISO 10993 biocompatible)
- Laptop host · USB-serial @ 115,200 baud · Arduino + TENS (CCFES concept)
- Python: NumPy, SciPy, Pygame, scikit-learn, Matplotlib

Methods

The vendor software only displays signals, so a custom decoder reads the raw two-byte serial stream directly. Each paradigm flashes timed stimuli, epochs the EEG around stimulus markers, baseline-corrects, and ensemble-averages to pull the tiny evoked response out of the noise.

1. **Brain**
   `Fpz electrode + mastoid reference`
2. **SpikerBox AFE**
   `$100 · 0.7–150 Hz · 10-bit ADC`
3. **USB serial**
   `custom binary decoder · ~488 Hz`
4. **DSP**
   `band-pass · epoch · baseline · average`
5. **Classify**
   `P300 amplitude in 250–500 ms window`
6. **Act**
   `polygraph · game · prosthetic trigger`

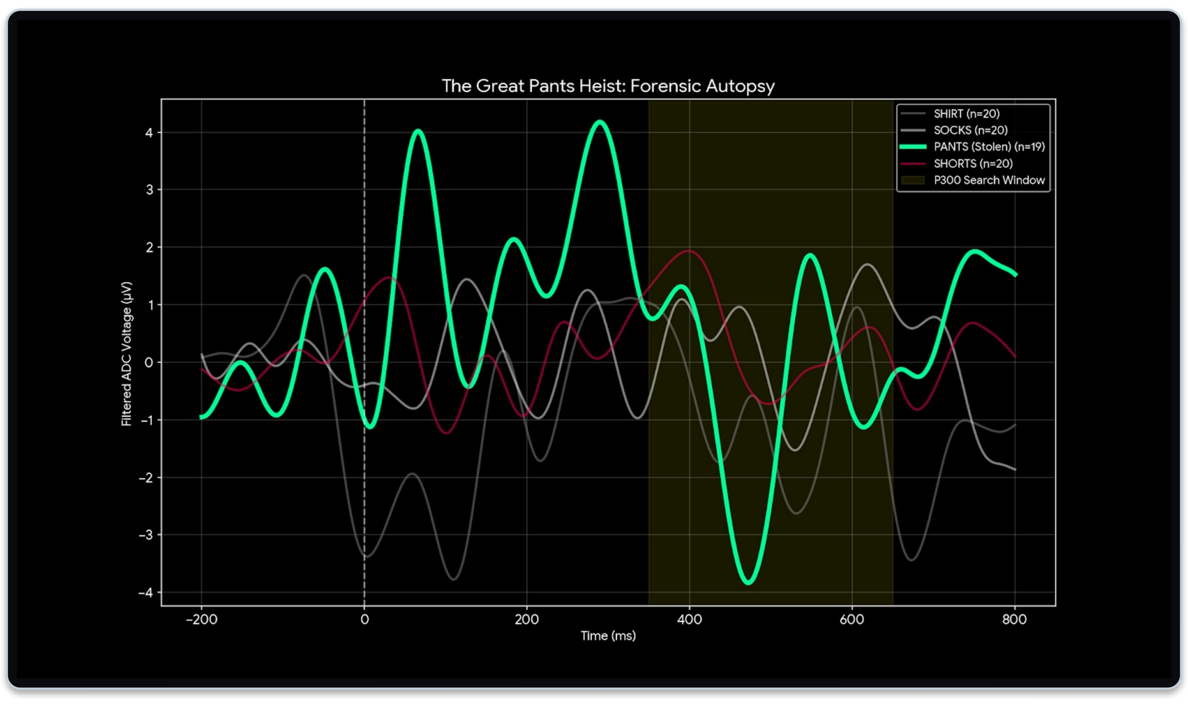
P300 Detection



Averaged response: the rare target (green) versus the frequent standard (grey). A 6.2 μV P300 peaks at 359 ms — the brain's recognition signal, isolated from a single $100 electrode by ensemble averaging.

Interpretation: the target's P300 stands clearly above the standard waveform, confirming that a cognitive ERP roughly two orders of magnitude smaller than an eye-blink can be recovered with averaging alone — no machine learning required.

Concealed-Information Test



A P300-based brain-polygraph. The secretly-counted stolen item (PANTS, green) produces the largest P300 spike while every innocent item stays below threshold — a clean 2:1 identification.

Results

Experiment	Result
Hardware + custom ECG	PASS
P300 oddball — 6.2 μV	PASS
CIT — Wallet theft	FAIL
CIT — Pants theft	PASS
CIT — Shirt theft	PASS
Face lineup — Epstein	PARTIAL
Face lineup — Trump	PASS
SSVEP (frontal site)	FAIL*
EOG / EMG game control	PASS
CCFES neuro-prosthetic	DESIGN

** expected failure — a single frontal electrode cannot see occipital (visual-cortex) signals.*

Discussion

Two of the failures were the most informative results. In the Wallet test the salient word "Diamond" drew a larger P300 than the true target — the Von Restorff novelty effect — so concealed-information tests must use semantically neutral stimuli. The frontal SSVEP null confirms electrode geography: occipital signals need occipital sensors. Both became reusable design rules.

Conclusion

A $100 single-electrode rig, driven by custom Python, reliably detects the P300 cognitive response and turns it into forensic, assistive, and rehabilitative applications. The ceiling on accessible neural interfacing is set by algorithms, not budget.

Future Work

- Add occipital (O1/O2) channels to unlock SSVEP & multi-paradigm BCI
- Train an ML classifier (SWLDA / SVM / CNN) on the data corpus
- Build and bench-test the CCFES drop-foot prosthetic

Acknowledgements & References

Advisor: Prof. Junaid Khan; WWU Department of Electrical & Computer Engineering. References: EECE 403 Final Design Report (2026); Backyard Brains SpikerBox Heart & Brain documentation.